Inertia, damping and sweep rate in two-dimensional dipolar compass-needle arrays

Scientific report V61: model, numerical methods, validation and results

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Code: `compass.py` 2.4.0 (Phase-1 fixes, quasi-static protocol with saddle check)
(git branch `fase2/dynamic-hysteresis`); usage: `USER_GUIDE.md`

Abstract

We study two-dimensional arrays of macroscopic compass needles: rigid planar rotors carrying point dipoles on fixed pivots, coupled by the full long-range dipolar interaction and obeying Newton's equation for rotation with viscous damping. The question is whether inertia and damping, summarized by an intrinsic quality factor Q_0, change collective reversal (hysteresis, avalanches, memory) in square, triangular and honeycomb arrays. An audit of the earlier simulations showed that their apparent Q dependence was dominated by numerical and protocol artefacts. A dipolar cutoff biased the interior field by 15–17 %. The time step was underresolved (1.4–1.8 % energy error at zero damping). Q depended on the field amplitude. Above all, the field sweep rate was uncontrolled: the loop area scales as $\sim(\text{rate})^{0.6}$. After correcting these issues, and with every correction covered by automated physics tests, we report two campaigns. (i) 690 hysteresis cycles at a controlled dimensionless sweep rate r (6×6 to 64×64 needles) show that for $Q_0 \lesssim 1$ the loops collapse when the rate is measured in units of the viscous relaxation time, and that at finite rate inertia narrows the loop by 20–50 %. (ii) 120 loops computed with a new event-driven quasi-static protocol, in which every rest state is checked against zero-torque saddles, show that in the limit $r \to 0$ the loop is set by the energy landscape in arrays up to 16×16: the gradient flow and $Q_0 = 1, 3, 10$ give the same area within 3.5 %, and $Q_0 = 1$ coincides with the gradient flow everywhere (within 10^{-3}). In 24×24 and 32×32 arrays, underdamped dynamics ($Q_0 = 3, 10$) narrows the quasi-static loop by 9–21 % in five of the six geometry/size combinations, through avalanches whose outcome differs from the gradient flow. The effect is not monotonic in size, and in the square lattice it varies between realizations. Inertia therefore matters strongly for *dynamic* hysteresis and, in the larger arrays, also selects reversal paths in the quasi-static limit; it is not a smooth modification of the loop.

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1 Introduction

Arrays of interacting magnetic dipoles are a classic model system for collective magnetism: frustration, domains, hysteresis and avalanche-like reversal arise from the long-range, anisotropic dipolar coupling alone. Macroscopic realizations with compass needles have been used to study dipolar ordering and thermodynamics [1] and the dependence of magnetization processes on the number and arrangement of needles [2]. Nanoscale realizations, artificial spin ice (ASI), have made frustration and its dynamics a large field [3, 4], including clocked and field-protocol-controlled dynamics [5], reversal via effective magnetic charges [6], relaxation pathways [7] and memory for computation [8].

The present system differs from ASI in a fundamental way. Nanoislands follow gyromagnetic (Landau–Lifshitz–Gilbert) dynamics and switch over shape-anisotropy barriers. A compass needle is a mechanical rotor with true moment of inertia, viscous (and in reality dry) friction, and no intrinsic switching barrier. ASI is therefore *motivation only*: no claim of equivalence is made.

Closest prior work and the gap. Macroscopic arrays of rotatable magnets with explicit inertia and dissipation have already been built and modelled. A planar lattice of magnetic rotors relaxes into a spin-ice phase over several time scales, and it was argued that microscopic ASI cannot capture inertial effects [9]. Chains of dipolar rotors have been used to design hysteresis, with the rotor time scale bounding the sweep rate below which the loop becomes rate-independent [10]. Separately, the dependence of the loop area on sweep rate (*dynamic hysteresis*) is a classic problem with scaling laws $A - A_0 \propto R^\beta$ [11]. What is missing is a controlled separation, in a long-range dipolar lattice, of three things: the quasi-static loop A_0, its dynamic excess $A - A_0$, and the role of inertia in each of them, across geometries and sizes. That separation is the subject of this report.

What this report contains. Section 2 defines the model and its scales; Sec. 3 the numerical methods, including the event-driven quasi-static protocol; Sec. 4 their validation. Section 5 quantifies the defects found in the earlier simulations; Sec. 6 reports dynamic hysteresis at controlled rate; Sec. 7 the quasi-static limit and the per-step avalanches. Section 8 discusses what is a rate effect and what is inertial; Secs. 9 and 10 list limitations and next steps. Software usage (flags, file formats, commands) is documented only in `USER_GUIDE.md`.

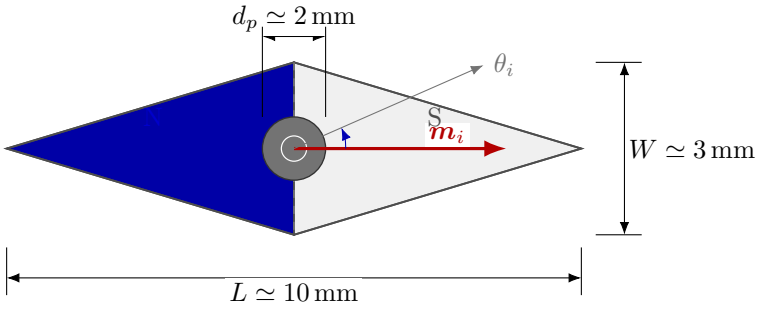
2 Model

2.1 Needle and lattices

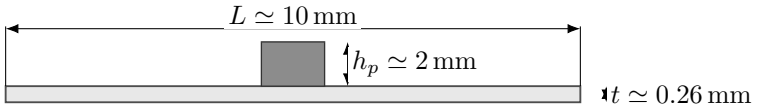
Each needle is a flat diamond-shaped steel lamina with a central pivot (Fig. 1). In the model it is reduced to a rigid planar rotor at a fixed lattice site $\boldsymbol{r}_i$ with one angular degree of freedom θ_i and moment $\boldsymbol{m}_i = m(\cos\theta_i, \sin\theta_i, 0)$. The blade shape is used only to compute $m = M_s\, t\, LW/2$ and the moment of inertia $I = \frac{1}{24} m_{\mathrm{blade}}(L^2 + W^2)$ (rhombus of length $L = 10\,\mathrm{mm}$ and width $W = 3\,\mathrm{mm}$; $M_s = 1.59 \times 10^6\,\mathrm{A/m}$, steel density $7850\,\mathrm{kg/m^3}$). The pivot spacing is 13 mm.

Open item: thickness. The drawing and the small prototype use $t = 0.26\,\mathrm{mm}$, whereas all simulations reported here use the engine default $t = 0.4\,\mathrm{mm}$. Since $m \propto t$ and $I \propto t$, the thickness rescales B_{ref} and the time unit but not the dimensionless results (Q_0, r, fields in units of B_{ref}). It must nevertheless be fixed against the real needle before comparison with experiment.

Three lattices are used, with the same nearest-neighbour distance r_{nn} (Figs. 2–4): square (coordination 4), triangular (6, the most frustrated) and honeycomb (3). All arrays are finite with free boundaries, as in the physical object.



Top view: two-colour diamond pointer with central pivot. The simulation keeps only $(\boldsymbol{r}_i, \theta_i)$ and $\boldsymbol{m}_i$; dimensions motivate I, m, and the optional pivot correction.



Side view: steel lamina and rectangular pivot cross-section, drawn with the same length scale and center as the top view.

Figure 1: Idealized geometry of the compass needle. The dynamics keeps only the pivot position and the angle; the blade dimensions set I and m. The drawing shows the prototype thickness $t \simeq 0.26\,\mathrm{mm}$; the simulations reported here use $t = 0.4\,\mathrm{mm}$ (see text).

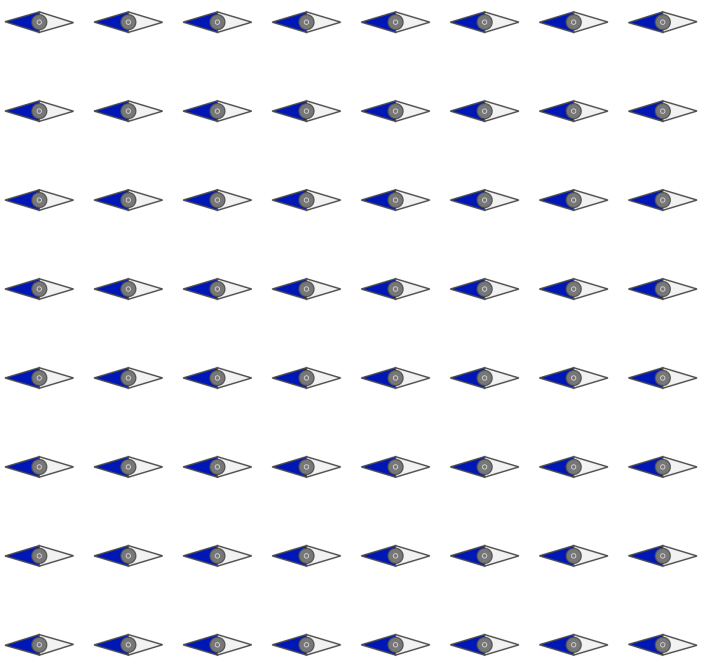


Figure 2: Square-lattice geometry used by `compass.py`. Pivots lie on a rectangular grid with four nearest neighbours in the bulk. This is the least frustrated of the three default geometries and provides the natural baseline for interpreting domains, hysteresis loops, and avalanche activity. Because the principal lattice directions are orthogonal, field sweeps along x or y have a simple relation to rows and columns of needles.

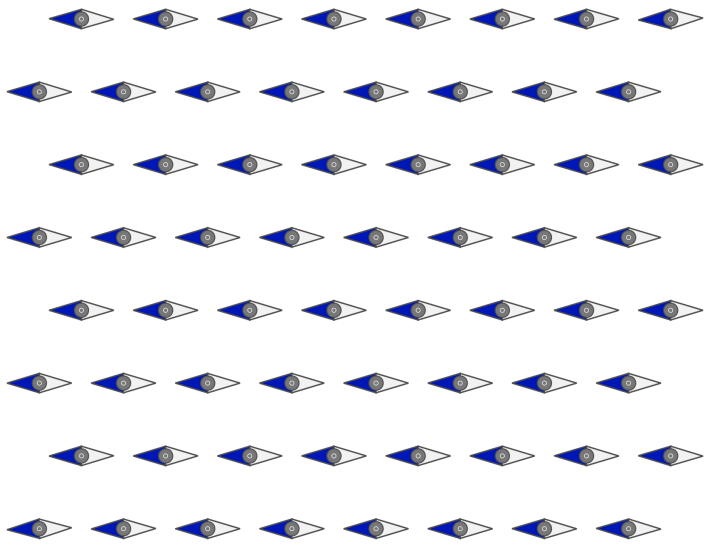


Figure 3: Triangular-lattice geometry. Each bulk pivot has six nearest neighbours, giving the highest local coordination among the default arrays. The equilateral triangular connectivity introduces strong geometric competition for dipolar alignment, so this geometry is expected to be especially sensitive to frustration, overshoot, and damping-controlled pathway selection. It is therefore a key test case for distinguishing purely static dipolar effects from genuinely dynamical inertial effects.

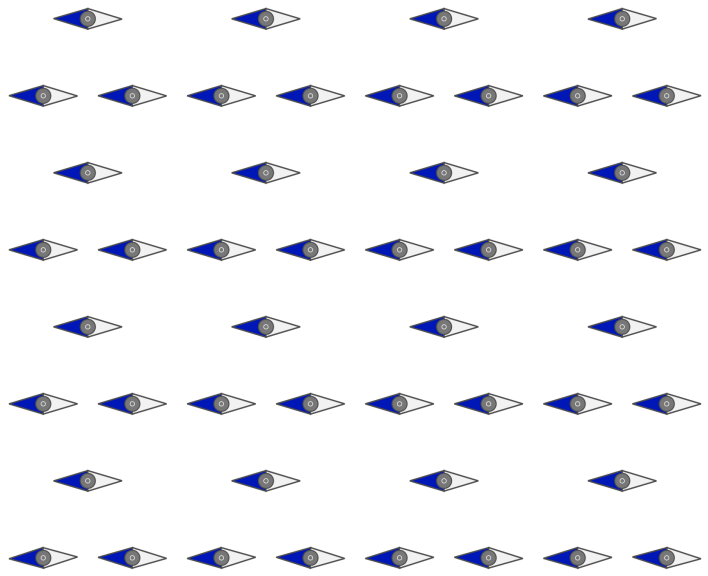


Figure 4: Honeycomb-lattice geometry. The honeycomb array can be viewed as a triangular construction with systematic removal of sites, producing hexagonal voids and three nearest neighbours per bulk pivot. This lower coordination changes both the available domain-wall paths and the local reversal constraints. It also gives a geometry closer in spirit to artificial-spin-ice networks, while retaining the continuous-angle compass-needle dynamics of the present model.

2.2 Interaction and dynamics

The field at needle i is the sum of the dipolar fields of all other needles and the applied field,

$$\boldsymbol{B}_i = \frac{\mu_0}{4\pi} \sum_{j \neq i} \frac{3(\boldsymbol{m}_j \cdot \hat{\boldsymbol{r}}_{ij})\hat{\boldsymbol{r}}_{ij} - \boldsymbol{m}_j}{r_{ij}^3} + \boldsymbol{B}_{\mathrm{ext}}(t), \qquad \tau_i = (\boldsymbol{m}_i \times \boldsymbol{B}_i)_z, \tag{1}$$

and each needle obeys

$$I\ddot{\theta}_i = \tau_i - b\,\dot{\theta}_i. \tag{2}$$

The model is deterministic ($T = 0$): randomness enters only through the initial angles. The energy is $E = E_{\mathrm{dip}} + E_{\mathrm{ext}} + E_{\mathrm{kin}}$ with $E_{\mathrm{dip}} = -\frac{1}{2}\sum_i \boldsymbol{m}_i \cdot \boldsymbol{B}_i^{\mathrm{dip}}$, $E_{\mathrm{ext}} = -\sum_i \boldsymbol{m}_i \cdot \boldsymbol{B}_{\mathrm{ext}}$ and $E_{\mathrm{kin}} = \frac{1}{2} I \sum_i \dot{\theta}_i^2$.

2.3 Scales: reference field, lattice field, quality factor and sweep rate

- $B_{\mathrm{ref}} = \frac{\mu_0}{4\pi}\frac{2m}{r_{\mathrm{nn}}^3}$ is a *normalization*: the on-axis field of one neighbour. Applied fields are quoted in units of B_{ref} (here $B_{\max} = 4B_{\mathrm{ref}}$).
- B_{lat} is the intrinsic field scale of the array: the median over sites of $|\boldsymbol{B}_i^{\mathrm{dip}}|$ in uniform states, averaged over directions. It is $\approx 2.2\,B_{\mathrm{ref}}$ for large square arrays, smaller for small arrays because of their edges.
- The quality factor $Q_0 = \omega_{\mathrm{lat}} I/b$ with $\omega_{\mathrm{lat}} = \sqrt{mB_{\mathrm{lat}}/I}$ is independent of the field protocol.
- The relaxation time is $\tau_{\mathrm{relax}} = \max(1/\omega_{\mathrm{lat}},\, b/(mB_{\mathrm{lat}})) = \max(1, 1/Q_0)/\omega_{\mathrm{lat}}$: inertial for $Q_0 \geq 1$, viscous for $Q_0 < 1$.
- The dimensionless sweep rate is $r = (dB/dt)\,\tau_{\mathrm{relax}}/B_{\mathrm{ref}}$: the field change, in units of B_{ref}, during one relaxation time. The quasi-static limit is $r \to 0$.

3 Numerical methods

Dipolar sum. The interaction is precomputed as a dense $K \times K$ tensor, and the field is a matrix–vector product. The default is the *full pair sum* over the finite array: a cutoff saves no time with a dense tensor and only biases the field (Sec. 5).

Integrator. Equation (2) is integrated with velocity Verlet, treating the damping implicitly:

$$\omega_i^{n+1} = \frac{\omega_i^n\left(1 - \frac{b\Delta t}{2I}\right) + \frac{\Delta t}{2I}\left(\tau_i^n + \tau_i^{n+1}\right)}{1 + \frac{b\Delta t}{2I}}. \tag{3}$$

The scheme is symplectic at $b = 0$ and stable for any b.

Time step. $\Delta t = f_{\mathrm{dt}}\,T_{\mathrm{loc}}$ with $T_{\mathrm{loc}} = 2\pi/\sqrt{mB_{\mathrm{loc}}/I}$. Here $B_{\mathrm{loc}} = m \max_i \max_{\hat{u}} \sum_j |A_{ij}\hat{u}| + \max_t |B_{\mathrm{ext}}|$ bounds the field any needle can feel in *any* configuration (≈ 6–$8\,B_{\mathrm{ref}}$). Then $\omega_{\mathrm{loc}}\Delta t \leq 2\pi f_{\mathrm{dt}} = 0.157$ for the default $f_{\mathrm{dt}} = 0.025$.

Energy ledger. The cumulative field work $W = -\sum_i \int \boldsymbol{m}_i \cdot d\boldsymbol{B}_{\mathrm{ext}}$ and dissipation $D = \int \sum_i b\,\dot{\theta}_i^2\,dt$ are integrated alongside (trapezoidal rule). They satisfy $\Delta E = W - D$ to $O(\Delta t^2)$, and over a closed field cycle W equals the loop area. The loop area is computed from the ledger, and cross-checked against $-\oint M\,dB$ to 5×10^{-3}.

Protocols. (a) *Rate sweeps*: the five-branch loop $0 \to +B_{\max} \to 0 \to -B_{\max} \to 0 \to +B_{\max}$ at a fixed dimensionless rate r; the area is measured on the closed cycle that starts at the first arrival at $+B_{\max}$. (b) *Event-driven quasi-static loop*: the same five branches in field steps $\Delta B = 0.005\, B_{\mathrm{ref}}$. After each step the dynamics runs at fixed field until rest, defined as $\max_i |\tau_i| \le 10^{-5} m B_{\mathrm{lat}}$ and $\max_i |\dot{\theta}_i| \le 10^{-5}\omega_{\mathrm{lat}}$. One rest state is recorded per step, so the number of reversed needles and the energy dissipated in that step define an avalanche. (c) *Gradient flow*: in the quasi-static protocol the relaxation can instead follow $b\dot{\theta}_i = \tau_i$, the $Q_0 \to 0$ limit without inertia.

Avalanche counters. A needle reverses when its projection on the drive axis changes sign. To ignore ringing, a Schmitt dead band of $\pm 30°$ around the perpendicular and a dwell time are required, so a ringing needle counts once per genuine reversal.

4 Validation

Every statement below is an automated test in `tests/test_physics.py`, run before every commit.

- The precomputed tensor equals the direct dipole–dipole pair sum to 10^{-12}, and the torque equals $-\partial E/\partial\theta_i$ (finite differences).
- At $b = 0$ the energy drift scales as Δt^2 (ratio ≈ 4 when Δt is halved). With damping, the energy never rises during free relaxation.
- The time-step bound dominates the actual fields for random and aligned states on all geometries, and an $8B_{\mathrm{ref}}$ hysteresis run is never flagged by the stability monitor.
- Q_0 is independent of the field amplitude and protocol, and $Q_0/Q_{\mathrm{ref}} = \sqrt{B_{\mathrm{lat}}/B_{\mathrm{ref}}}$.
- The ledger balance holds to $O(\Delta t^2)$ (the residual ratio is ≈ 4 when Δt is halved), and the closed-cycle field work equals the loop area.
- The loop area converges as $r \to 0$ on a small array.
- Quasi-static protocol: logged states are mechanical equilibria within the tolerances; the ledger balance includes the field jumps; and the gradient flow reproduces the low-Q_0 inertial loop.

Tolerance of the quasi-static protocol. The rest criterion matters. On a 6×6 square array at $Q_0 = 0.3$, a loose tolerance (10^{-3}, $\Delta B = 0.02 B_{\mathrm{ref}}$) gives $a_0 = 0.710$, because relaxation stops early near instabilities, where the dynamics slows down critically. The validated settings (10^{-5}, $\Delta B = 0.005 B_{\mathrm{ref}}$) give $a_0 = 0.461$ for inertial dynamics and 0.4606 for the gradient flow. The slowest rate sweep ($r = 3 \times 10^{-4}$) gives 0.473.

Saddle check. A state with zero torque and zero velocity can be a saddle of the energy rather than a minimum. The gradient flow, in particular, preserves the symmetry of the array and can stay on a symmetric, unstable branch indefinitely. Each rest state is therefore perturbed by random angular kicks of 10^{-4} rad and relaxed again before it is recorded; the kick energy is booked in the ledger. A first quasi-static grid run without this check produced spurious differences between dynamics of up to 43 % in small honeycomb arrays, which disappear with it. Over the 120 loops reported here, 109 of the 480 000 field steps (0.02 %) reached the relaxation-time cap before the tolerance, almost all (100) in gradient-flow or $Q_0 = 1$ runs on $N \ge 16$ arrays, where very soft modes relax slowly.

5 Results I: what was wrong in the earlier simulations

The figures of the July 2026 campaign (loop area and avalanche exponent versus Q) were affected by four defects, quantified on the corrected code:

1. **Cutoff.** A cutoff of $3.5\,r_{\mathrm{nn}}$ underestimated the interior field of a uniformly magnetized array by 15 % (20 × 20) to 17 % (36 × 36), and removed edge and shape fields entirely.

2. **Time step.** Setting Δt from the single-neighbour field left $\omega_{\mathrm{loc}}\Delta t \approx 0.5$. The energy error at $b = 0$ was 1.8 % over 2 s of a 12 × 12 array, against 0.14 % with the corrected step.

3. **Quality factor.** Q was built from $\max(B_{\mathrm{ref}}, B_{\mathrm{ext}})$, so the same damping gave a $\sqrt{8} \approx 2.8$ times larger Q in an $8B_{\mathrm{ref}}$ loop than in a static run. The campaign script used yet another convention.

4. **Sweep rate.** With t_{sim} fixed, the rate was uncontrolled. On a 12 × 12 array the loop area fell roughly as $(\mathrm{rate})^{0.6}$ at every Q and never reached a quasi-static value. The old area-versus-Q curve had the shape of a viscous (Debye-like) lag, not of collective physics.

The avalanche exponents of the earlier analysis rested on tails of 30–40 events and are not retained.

6 Results II: dynamic hysteresis at controlled sweep rate

Campaign. 600 cycles cover three geometries, $N = 6, 10, 16, 24, 32$, $Q_0 = 0.1, 0.3, 1, 3, 10$ and $r = 10^{-1}$ to 10^{-3} (down to 3×10^{-4} for $N \le 16$), plus seed checks. A GPU extension adds 90 cycles at $N = 48, 64$ ($Q_0 = 0.3, 1, 3$; $r \ge 10^{-3}$). Damping was set per geometry and size so that Q_0 is exact. All 690 runs passed quality control: balance residual $\le 1.2 \times 10^{-3}$, no stability flags, and $M \ge 0.995$ at $B_{\max}$.

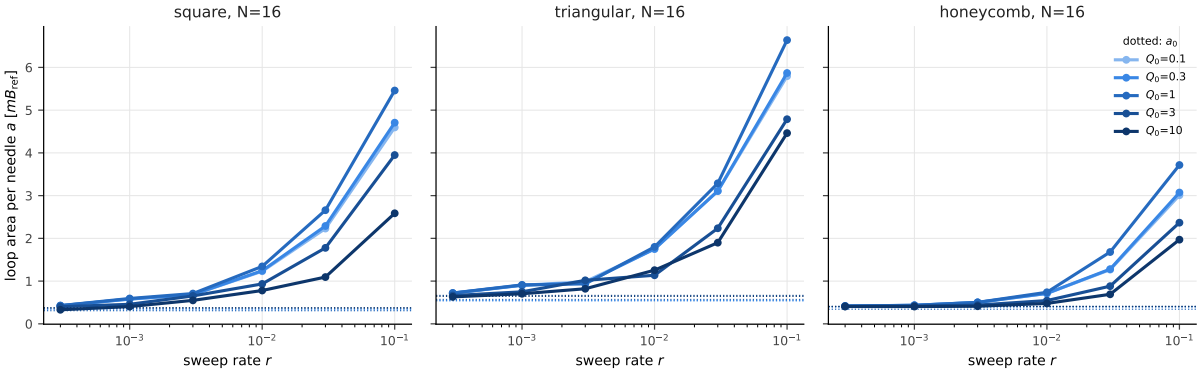


Figure 5: Loop area per needle a (units of mB_{ref}) versus dimensionless sweep rate r, $N = 16$. The curves for $Q_0 = 0.1$, 0.3 (and largely 1) coincide: with r measured in units of the viscous relaxation time, overdamped dynamics is a pure rescaling of time. Underdamped arrays ($Q_0 = 3, 10$) have narrower loops at the same r. Dotted lines: fitted a_0, unreliable for the larger arrays (see text).

Overdamped collapse. For $Q_0 = 0.1$ and 0.3 the areas differ by at most 2.4 % over all geometries, sizes and rates (Fig. 5). In the overdamped limit Eq. (2) reduces to $b\dot{\theta} = \tau$, and b only rescales time. This is a control result: any apparent Q dependence for $Q \lesssim 1$ at fixed t_{sim} is a rate effect.

Inertial narrowing at finite rate. At the same r, raising Q_0 from 0.1 to 10 reduces the area by 22–44 % at $r = 10^{-1}$ and by 28–54 % at $r = 10^{-2}$, depending on geometry and size. Underdamped loops show ringing and discrete jumps (Fig. 6).

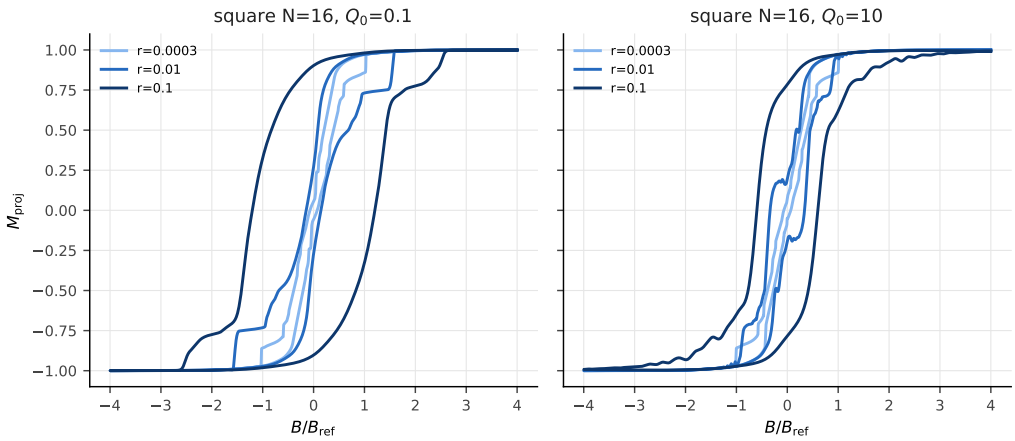


Figure 6: Hysteresis loops $M_{\mathrm{proj}}(B)$ of a 16×16 square array at three sweep rates, overdamped ($Q_0 = 0.1$, left) and underdamped ($Q_0 = 10$, right). Faster sweeps widen the loop; underdamped loops show ringing and discrete avalanche jumps.

The quasi-static limit is out of reach by sweeping. For $N \geq 24$ the area still changes by 40–98 % between $r = 3 \times 10^{-3}$ and 10^{-3}, and extrapolations $a(r) = a_0 + c\, r^{\beta}$ are not reliable (a_0 hits zero). Reaching $r \sim 10^{-5}$ by sweeping is prohibitive, since the cost grows as $1/r$, and faster still for $Q_0 < 1$. This motivated the quasi-static protocol of Sec. 7.

Size dependence at fixed rate. At $r = 10^{-3}$ the area levels off for $N \geq 32$. For $Q_0 = 0.3$: square $0.265 \to 0.238 \to 0.235$ for $N = 32, 48, 64$; honeycomb ≈ 0.42; triangular 0.73–0.69. Below $N \approx 24$ the dependence is non-monotonic, reflecting edges and, for the triangular lattice, the shape of the cut.

Seed sensitivity in sweeps. With uniform damping the saturated state erases the initial state in most conditions. In underdamped honeycomb arrays, however, the area varied by up to 27 % between seeds. Section 7 shows that this is a dynamic effect.

7 Results III: the quasi-static limit and per-step avalanches

Campaign. 120 quasi-static loops cover three geometries, $N = 6, 10, 16, 24, 32$, four relaxation dynamics (gradient flow, $Q_0 = 1, 3, 10$) and two seeds, with 4000 field steps each and the saddle check of Sec. 4. The ledger residual was $\leq 4 \times 10^{-4}$ for the inertial runs and $\leq 8.4 \times 10^{-3}$ for the gradient flow (whose Euler step carries a small kinetic-energy bookkeeping term). The area from the ledger exceeds the staircase quadrature of $-\oint M\, dB$ by $0.010\, m B_{\mathrm{ref}}$ in every run, the offset expected from recording M after each step of $\Delta B = 0.005 B_{\mathrm{ref}}$.

$Q_0 = 1$ is gradient flow. In all 15 geometry/size combinations the $Q_0 = 1$ loop coincides with the gradient-flow loop within 10^{-3}, including the realization-to-realization variation of the square 10×10 array (below). A critically damped rotor does not overshoot, so it follows the same sequence of basins as the gradient flow.

Small arrays: no inertial effect. For $N \leq 16$ the area changes with Q_0 by less than 1 % in triangular and honeycomb arrays, apart from honeycomb 6×6 at $Q_0 = 10$ (-2.9 %). In square arrays the $Q_0 = 10$ loop differs by $+2.7$ % (10×10) and $+3.5$ % (16×16), within the seed-to-seed variation of those conditions.



Figure 7: Quasi-static loop area a_0 versus array size for the four relaxation dynamics (bars: range over two seeds). Up to $N = 16$ all dynamics give the same area; at $N = 24$ and 32 the underdamped loops ($Q_0 = 3, 10$) are narrower in five of the six geometry/size combinations. The size dependence is non-monotonic below $N \approx 24$ (edges, and for the triangular lattice the shape of the cut).

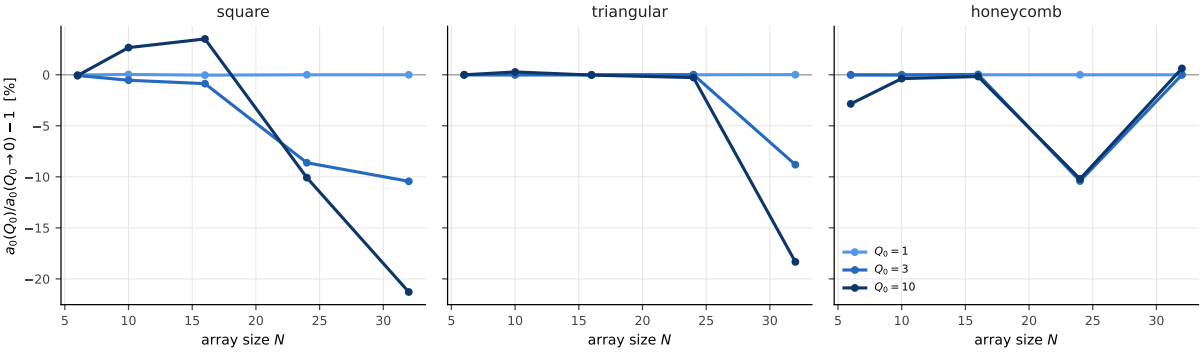


Figure 8: Inertial effect on the quasi-static loop: $a_0(Q_0)/a_0(Q_0 \to 0) - 1$ versus array size (mean over two seeds).

Large arrays: inertia selects narrower loops. At $N = 24$ and 32 the underdamped loops are narrower than the gradient-flow loop in five of the six combinations (Fig. 8): square 24×24 ($Q_0 = 3$: $-9\,\%$, $Q_0 = 10$: $-10\,\%$), square 32×32 ($-10\,\%$, $-21\,\%$), triangular 32×32 ($-9\,\%$, $-18\,\%$) and honeycomb 24×24 ($-10\,\%$ for both). Triangular 24×24 and honeycomb 32×32 show no effect ($\leq 0.6\,\%$). The sign is always the same: when inertia changes the loop, it narrows it. The coercive field and remanence change along with the area; for triangular 32×32 at $Q_0 = 10$ the remanence drops from 0.22 to 0.07. These changes exceed the ledger and discretization uncertainties by two orders of magnitude. The comparison with the first quasi-static grid, run without the saddle check, shows why the check matters here too: the gradient-flow loops of square 24×24, square 32×32 and triangular 32×32 were 13–20 % wider there, because the flow stayed on marginally unstable states, while the honeycomb 24×24 values are unchanged. The differences quoted above are measured against the corrected gradient flow, in which no state is a saddle.

The pattern points to discrete path selection. During a few large avalanches the overshoot of the rotors carries the array into a different basin than the gradient flow would, and the subsequent branch of the loop changes. Because it depends on a few events, the effect is not monotonic in N and depends on geometry. Larger arrays have more and larger avalanches, which makes such events more likely, but two sizes per geometry are not enough to establish a trend.

Realization dependence. In triangular and honeycomb arrays the two seeds give identical loops (spread $\leq 10^{-3}$). In square arrays they do not, in seven conditions: the gradient flow and $Q_0 = 1$ at 10×10 (spread 14 %), and $Q_0 = 3$ or 10 at $N \geq 10$ (7–23 %). For example, square 32×32 at $Q_0 = 3$ gives $a_0 = 0.152$ for one seed and 0.191 (equal to the gradient flow) for the other. With the saddle check, the seed also sets the direction of the random kicks, so at a symmetric branch point it chooses which branch the array takes. The square lattice, with its degenerate reversal paths, is the geometry where this matters. The seed sensitivity of the rate sweeps (honeycomb, Sec. 6) does not survive in the quasi-static limit and is a finite-rate effect. Two seeds only bracket the realization dependence; its distribution needs more realizations.

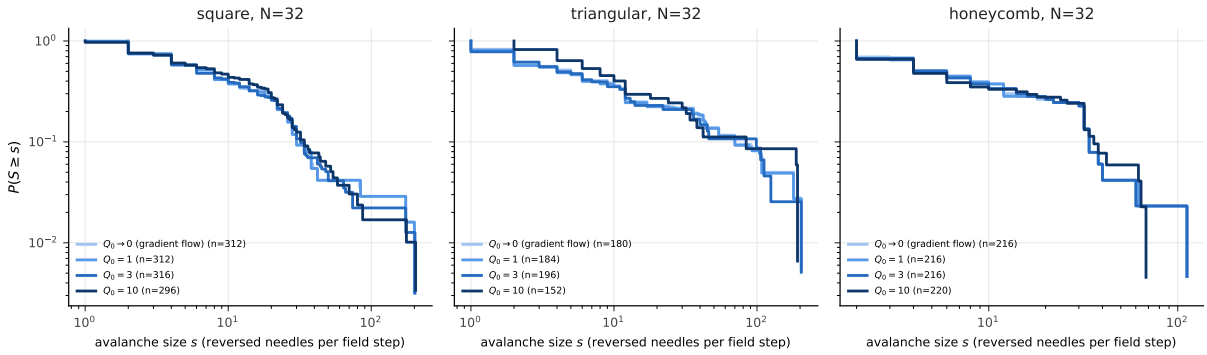


Figure 9: Complementary cumulative distribution of the avalanche size s (needles reversed per field step) for $N = 32$ and the four dynamics, over the closed cycle (the first branch, which starts from the random initial state, is excluded). The distributions are similar for all dynamics; the largest events involve ≈ 200 needles (20 % of the square and triangular arrays).

Avalanches. Over the closed cycle the per-step avalanche-size distributions are broad and cut off by system size (largest events ≈ 200 needles at $N = 32$). They are similar for all dynamics (Fig. 9), with 150–320 events per condition (two loops) at $N = 32$. Events larger than the number of needles occur only during the first relaxation from the random initial state, not during the cycle. With these sizes no scaling exponent is claimed.

8 Discussion

The picture that emerges separates cleanly into three parts:

1. **Overdamped regime ($Q_0 \lesssim 1$): no dependence on damping.** The damping is not a parameter of the physics, only of the time scale. With r measured in viscous units the loops collapse at every rate, and in the quasi-static limit $Q_0 = 1$ already coincides with the gradient flow.

2. **Dynamic hysteresis: large inertial effects.** At finite rate inertia reduces the loop area by tens of per cent and makes the loop depend on the initial state (honeycomb). In this sense inertia is a genuine control parameter of the *dynamic* response, consistent with the rotor time scale bounding rate-independent hysteresis in dipolar chains [10] and with the general phenomenology of dynamic hysteresis [11].

3. **Quasi-static loop: inertia selects paths in large arrays.** In the limit $r \to 0$ the energy landscape fixes the loop in arrays up to 16×16, whatever the damping. In 24×24 and 32×32 arrays underdamped dynamics narrows the loop by 9–21 % in most cases, through discrete events in which overshoot changes the outcome of an avalanche. The effect depends on geometry and size, and in the square lattice also on the realization.

The third point connects naturally to *memory*: whether a system returns to the same state after a field excursion (return-point memory) is guaranteed for overdamped ferromagnetic-like dynamics [12], but inertial path selection is exactly the kind of mechanism that can break it. Memory formation in driven disordered matter is an active field [13], and the compass array offers a mechanical, fully characterized platform to test it. This is the proposed Phase 3.

Compared with the rotor spin-ice study of Ref. [9], the present work adds a controlled separation of quasi-static and dynamic contributions across three geometries and sizes, a protocol-independent definition of Q_0, a quasi-static protocol that rejects saddle points, and the observation that inertia acts on the quasi-static loop of large arrays through discrete path selection rather than smoothly.

9 Limitations

- **Point dipoles.** The needle length is 0.77 of the pivot spacing, so near-field (multipolar) corrections are not small. A dumbbell (end-charge) model is needed as a robustness check.

- **Viscous damping only.** Real pivots have dry (Coulomb) friction, which acts as a switching threshold and quenched disorder. It is absent here.

- **$T = 0$, deterministic.** There is no thermal bath. Disorder enters only through the initial state and the saddle-check kicks; triangular and honeycomb quasi-static loops forget both, square loops do not always.

- **Finite size and edges.** Arrays up to 32×32 (quasi-static) and 64×64 (sweeps), all with free boundaries. Below $N \approx 24$ the size dependence is non-monotonic. The inertial effect of Sec. 7 appears only at $N \geq 24$; a denser size scan, larger arrays and more seeds (or weak disorder) are needed to establish its statistics and its size dependence.

- **Quasi-static protocol.** Finite ΔB and tolerance: changing ΔB from 0.01 to $0.005 B_{\mathrm{ref}}$ changes a_0 by 2–3 % on 6×6 arrays, and 0.02 % of the steps hit the relaxation cap.

- **Thickness** not yet fixed against the real needle (dimensionless results unaffected).

10 Next steps

1. **Inertial path selection.** Locate the avalanches whose outcome differs between gradient flow and $Q_0 = 10$ in the 24×24 and 32×32 arrays (same field step, different final state), and map them in space. Then establish their frequency and size dependence with $N = 40$–64 (GPU) and more seeds, using weak damping disorder to sample realizations.

2. **Memory (Phase 3).** Nested subloops and FORC in the quasi-static protocol: does return-point memory fail with inertia, and in which geometry?

3. **Dynamic excess.** $a(r) - a_0$ with a_0 from the quasi-static protocol: test rate scaling and collapse across Q_0 and geometries.

4. **Robustness.** Dumbbell interaction, Coulomb friction as controlled disorder, sample shape.

5. **Experiment.** Measure I, m, b and the static friction of real needles; compare loops at two or three rates and a memory protocol on a small array.

A Reproducibility

- Code: https://github.com/jpsinnecker/compass, branch `fase2/dynamic-hysteresis`. Engine `compass.py`; campaigns `fase2_campaign.py` (`-protocol sweep|qs`, sharding across machines); analysis `fase2_analysis.py` and `fase2_qs_analysis.py`; tests `tests/`.

- Runs: Mac (Apple M2 Max, CPU) and a Linux workstation (Intel i9-9900K CPU, NVIDIA RTX 3090 GPU for $N \geq 48$). The job lists of each shard were checked to be identical on both machines.

- Data (outside git): `~/compass_runs/fase2*`. Each run directory has a CSV time series, JSON metadata (configuration, derived scales, energy ledger, quality-control fields) and initial/final state files.

- Commands, flags and file formats: `USER_GUIDE.md` (PDF: `docs/USER_GUIDE.pdf`).

B Condensed history

June–July 2026: visual simulator, geometries, field protocols, precomputed tensor (V40), FORC and demagnetization (V67–V79), research branch with metadata and observables (V2.0.2), ringing-proof avalanche counters and stability monitor (V2.1). Reports V51–V53. October 2026: physics audit; full pair sum, time step from a local-field bound, intrinsic Q_0, energy ledger, dimensionless sweep rate (Phase 1, with physics tests); dynamic-hysteresis campaign and event-driven quasi-static protocol with saddle check (Phase 2; Reports V60–V61). Full detail: git history and `docs/AUDIT_PHYSICS_ROADMAP.md`.

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